Angular Selection Rules and Time-Delay Relations for Berger–Hopf Lensing in a Closed Cosmology

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Abstract

We calculate angular light transfer and linked strong-lensing time delays in a conditional effective realization of the Berger–Hopf framework. The calculation retains the full nine-dimensional scalar $n = 2$ harmonic space of a closed spatial geometry and six physical initial coordinates for each harmonic. Its observer-centered angular decomposition contains one monopole, three dipoles and five quadrupoles. Only the quadrupoles source first-order smooth tracefree beam deformation. We combine their normalized spatial Hessians with an action-derived temporal propagator to obtain the observer–lens, observer–source and lens–source response matrices. A specified quadrupole has a nodal central ray with vanishing first-order frequency and trace responses but nonzero tidal deformation. For a fixed physical singular isothermal lens and source displacement, the resulting relative-delay correction is proportional to $\cos(2\varphi)$ and to a difference of two propagation kernels. An illustrative normalized matter-density coefficient of 10^{-3} gives a -0.353216 s shift at $(z_l, z_s) = (0.3, 1)$ along one screen axis, the opposite shift after a 90° source rotation, and a first-order null at 45°. Independent angular-ray and stationary-image calculations check the result. We give an exact effective-potential representation and a restricted source-redshift consistency contrast. These are reproducible conditional predictions; neither an observed amplitude nor a unique microscopic state is selected.

Keywords: strong gravitational lensing; time delays; closed cosmology; scalar harmonics; Jacobi maps; Berger–Hopf framework

1 Introduction

Strong-lensing calculations connect a physical preparation of geometry and matter to image positions, beam distortions and arrival times. In the Berger–Hopf framework used here, the lensing encounter is interpreted as light traversing an envelope together with its environmental state. A useful calculation must therefore specify one common smooth environmental state for the incoming and outgoing propagation, together with the physical local encounter. Here the local lens is separately prescribed and held fixed. The calculation must determine which optical quantities this joint preparation fixes. This paper develops such a conditional forward calculation for an angular sector that supports nonzero tidal light transfer.

The retained effective realization, denoted R1 in the accompanying source archive, is a cubic scalar–tensor action coupled to dust and radiation on a closed homogeneous background [1, 2]. We use its six-dimensional linear initial-value problem to supply the temporal metric responses. We then extend the optical calculation to the entire scalar $n = 2$ spatial eigenspace. This gives a precise question: *which angular beam distortions and relative arrival-time changes must accompany*

one specified preparation of this state? The answer is a family of normalized response coefficients and, for one explicit preparation, a nonzero timing law.

The mathematical ingredients have established precedents. Harmonics on the three-sphere provide the spatial mode decomposition [3]; scalar shear is a spin-two optical response [4]; and general distance-matrix lens theory supplies the physical arrival-time function [5, 6]. The contribution here is their explicit composition with the declared six-state dynamics, the calculated redshift-dependent coefficients, and the linked angular shear–delay prediction. The monopole/dipole selection rule and the generic matrix Fermat construction are used as established geometry, with derivations included to fix normalization and signs.

The interpretation also requires care. A calculated envelope response can have an ordinary effective lens-potential representation. Such a correspondence is compatible with the BHSM interpretation of the same surface contributions. We construct that representation explicitly; we do not require a discrepancy from conventional lensing for the forward calculation to be meaningful. Conversely, a nonzero conditional response does not by itself identify the physical state from observed images. Mass-sheet and source-position freedoms remain relevant [6, 7].

This is a separate theoretical and computational study. Its inputs are the stated effective action, frozen homogeneous background, normalized initial coordinates and prescribed thin-lens encounter. No quasar photometry, telescope pixels, astrometric fit, detector calibration or observationally fitted state amplitude enters the calculation.

2 Physical preparation and approximation

2.1 Effective dynamics and state coordinates

With signature $(-,+,+,+)$, the effective action is

$$S=\int d^4x\sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}\mathcal{R}-\frac{1}{2}(\nabla T)^2+\frac{(\nabla T)^2\Box T}{\Lambda^3}-V(T)+\mathcal{L}_m+\mathcal{L}_r\right],\qquad V(T)=V_0-\mu^3 T. \tag{1}$$

The cubic interaction belongs to the established kinetic-gravity-braiding and generalized scalar–tensor action class [8, 9]. Its form is not claimed as a new interaction in this work. Here T is the scalar field, $\mathcal{R}$ the spacetime Ricci scalar, and the two matter sectors are pressureless dust and radiation. The symbol T used later for the time-delay *matrix* is always distinguished by its matrix equation and context. The numerical provider uses $M_{\rm Pl}=H_0=c=1$, with rescaled cubic parameter $\lambda=\Lambda^3/(M_{\rm Pl}H_0^2)$ and slope $q_T=\mu^3/(M_{\rm Pl}H_0^2)$. Its parameters are

$$\Omega_{m0}=0.31,\quad \Omega_{r0}=9\times10^{-5},\quad \Omega_{k0}=-0.018,\quad K=0.018,\quad \lambda=1,\quad q_T=3. \tag{2}$$

The frozen background uses $V_0=2.3986073449708067$ in $M_{\rm Pl}^2H_0^2$ units and the archived homogeneous initial conditions; it is reused without tuning to any lens. Appendix A describes those conditions and the quadratic-action reduction.

Each real normalized $n=2$ harmonic carries the physical initial vector at $z_i=2.1$,

$$y=\left(\zeta,\Pi_{2,\rm ref},\delta_m^{\rm ADM},\delta_r^{\rm ADM},v_m^{\rm ADM},v_r^{\rm ADM}\right)_i^T. \tag{3}$$

The labels specify the gauge and normalization of the input coordinates; they are not six independently fitted properties of a particular lens. The reference momentum $\Pi_{2,\rm ref}$ is converted to the full canonical state together with the other five inputs. It is not identified with the canonical momentum of the coupled scalar–matter system. Rotational symmetry of the homogeneous background makes the same temporal propagator applicable to every member of the $n=2$ eigenspace. It does not select their angular coefficients or a preferred axis.

This effective action is a declared conditional realization within the BHSM research program. The paper does not derive it as a unique microscopic closure or compute a unique normalized local envelope state. The physical content of the results is consequently of the form: *given this effective action and this initial preparation, these optical responses follow.* The explicitly retained dust and radiation coordinates permit environmental perturbations to participate in that prediction.

2.2 Optical experiment

Lengths in the field and propagation calculation are measured in c/H_0. The normalized harmonics and their coefficients use that same coordinate unit; numerical comparisons of different state coordinates therefore refer to the declared basis. A value $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is used only to turn the illustrative distances, fixed physical source displacement and delays into conventional units. All lens/source redshifts in the examples lie below the initial surface at $z_i = 2.1$.

We retain terms linear in the smooth scalar perturbations and quadratic in paraxial transverse separations. A local weak-field thin lens supplies the strong-lensing encounter. "Strong" describes the multiple imaging; it does not require a strong local metric perturbation. The local potential is held fixed when the smooth environmental preparation is varied. In the numerical example it is a singular isothermal sphere (SIS), prescribed as an effective physical lens. Its microscopic source and a fully matched nonlinear body solution are not derived here.

The observable delay is measured by the observer's clock at fixed measured lens and source redshifts. The comparison also fixes the transverse *physical* source displacement and the physical lens strength. This choice is part of the experiment: fixing the angular source position instead changes the derivative. The smooth background Jacobi maps describe one common narrow beam; spatial variation across a finite field beyond the retained paraxial order is outside the calculation. All maps use one parallel-transported screen and nonconjugate vertex pairs. The measured redshifts specify the reference vertices; no image-specific endpoint retuning is performed. The SIS is a local encounter model, rather than a global finite-mass galaxy solution, and its structural reaction to a changing environment is outside this fixed-body experiment.

3 Normalized angular sectors and optical selection

Let $R = K^{-1/2}$, $u = \chi/R$, and $S_K(\chi) = R\sin u$. The spatial metric is $\gamma = d\chi^2 + S_K^2 d\Omega^2$. Write the unit-sphere embedding coordinates as $X^0 = \cos u$, $X^i = \sin u\, n^i$. For a real symmetric tracefree 4×4 matrix A, define

$$Q_A = \mathcal{N} X^I A_{IJ} X^J, \qquad \mathcal{N} = \sqrt{\frac{6}{\pi^2 R^3}}. \tag{4}$$

The ambient polynomial is homogeneous harmonic of degree two; hence $D^2 Q_A = -8KQ_A$. The fourth moment $\langle X_I X_J X_K X_L\rangle = (\delta_{IJ}\delta_{KL} + \delta_{IK}\delta_{JL} + \delta_{IL}\delta_{JK})/24$ gives

$$\int_{S^3(R)} Q_A Q_B \, dV_\gamma = \mathrm{Tr}(AB). \tag{5}$$

A Frobenius-orthonormal matrix basis therefore supplies all nine normalized $n = 2$ harmonics. Decomposing $A_{ij} = C_{ij} - A_{00}\delta_{ij}/3$, $\mathrm{Tr}\, C = 0$, yields

$$\begin{aligned} Q_A/\mathcal{N} = {} & A_{00}(\cos^2 u - \tfrac{1}{3}\sin^2 u) + 2A_{0i}\sin u\cos u\, n^i \\ & + \sin^2 u\, n^i C_{ij} n^j. \end{aligned} \tag{6}$$

These are the $1+3+5$ dimensions of the observer-centered angular sectors $\ell = 0, 1, 2$. The spatial harmonic index n on S^3 and the angular index ℓ on the observer's sky are distinct.

For a screen E satisfying $E^TE = I_2$, $E^Tn = 0$, the tracefree spatial Hessian of $f(\chi)Y_{\ell m}$ is

$$(D_{\hat{A}}D_{\hat{B}}[fY])_{\rm TF} = \frac{f}{S_K^2}\left(\nabla_A\nabla_B Y - \frac{1}{2}s_{AB}\Delta_\Omega Y\right). \tag{7}$$

The radial connection contribution is proportional to I_2. For $\ell = 1$, $\nabla_A\nabla_B Y = -s_{AB}Y$ and $\Delta_\Omega Y = -2Y$, so the bracket vanishes. The $\ell = 0$ bracket also vanishes. Thus the four monopole/dipole spatial modes do not source first-order smooth cosmological shear on observer-centered radial background rays. This statement does not remove their trace, redshift or timing responses, and does not remove shear produced by the galaxy encounter.

For a normalized quadrupole $Q_C = \mathcal{N}\sin^2 u\, n^TCn$ with $\mathrm{Tr}(C^2) = 1$, let

$$H_C(n) = 2E^TCE + (n^TCn)I_2. \tag{8}$$

This matrix is symmetric and tracefree. The exact cancellation of the radial factor gives

$$(D_{\hat{A}}D_{\hat{B}}Q_C)_{\rm TF} = \mathcal{N}KH_C(n). \tag{9}$$

4 State-derived tracefree propagation

In the convention $ds^2 = a^2[-(1+2\Phi)d\eta^2 + (1-2\Psi)\gamma_{ij}dx^idx^j]$, write the current six temporal metric responses as $\phi_j(\eta)$ and $\psi_j(\eta)$, and let $w_j = \phi_j + \psi_j$. The initial coordinates are $(\zeta, \Pi_{2,\rm ref}, \delta_m^{\rm ADM}, \delta_r^{\rm ADM}, v_m^{\rm ADM}, v_r^{\rm ADM})$ for each normalized spatial harmonic. No equality between the two metric potentials is imposed.

The background physical angle-to-separation map between vertices a, b is $d_{ab} = a_bS_K(\chi_b - \chi_a)$. The physical initialization $D'_a = \omega_aI_2$, with one observer-normalized affine parameter, becomes a unit conformal derivative at χ_a. Linearizing the Sachs equation gives

$$(\partial_\chi^2 + K)\delta\widetilde{D}_{ab}^{\rm TF} = -S_K(\chi - \chi_a)[D_\perp D_\perp(\Phi + \Psi)]_{\rm TF}. \tag{10}$$

Define shear by $\delta D_{ab}^{\rm TF} = -d_{ab}\Gamma_{ab}$. For an orthonormal STF three-matrix basis C_A, $A = 1, \ldots, 5$, the result is

$$\Gamma_{ab}(n) = \sum_{j=1}^{6}\sum_{A=1}^{5} g_{ab,j}y_{jA}H_{C_A}(n), \tag{11}$$

$$g_{ab,j} = \frac{\mathcal{N}K}{S_K(\chi_b - \chi_a)}\int_{\chi_a}^{\chi_b} S_K(\chi_b - \chi)S_K(\chi - \chi_a)w_j(\eta_o - \chi)\, d\chi. \tag{12}$$

The three maps ol, os, and ls are independently initialized. The final scale a_b cancels from this fractional shear, and no additional redshift multiplier belongs in the kernel.

At fixed measured redshift, endpoint motion and moving-vertex terms multiply the background identity at first order. Corrections to a first-order shear integral from those shifts are second order. Local aberration is conformal on an infinitesimal celestial sphere and supplies no first-order spin-two term. These facts establish the tracefree result; they do not replace the frequency, radial-event and local-scale terms required for a generic trace response.

The full retained state has 54 coefficients, of which at most 30 enter this smooth-shear response. For redshift/vertex labels r, the five angular coefficients satisfy $S_{rA} = G_{rj}y_{jA}$, $G_{rj} = g_{r,j}$. A left-null vector of the actual kernel design therefore supplies a shared-state relation. Its observational relevance depends on conditioning, units and the specified lens/source nuisance model. If all six state components share a single spatial quadrupole C, then $\Gamma_{ab} = (g_{ab}\cdot y)H_C$, and

$$\gamma_{1,ab}\gamma_{2,cd} - \gamma_{2,ab}\gamma_{1,cd} = 0. \tag{13}$$

This profile restriction is not automatic for a generic six-by-five quadrupole preparation, and the deterministic identity is not a stochastic covariance prediction.

5 A complete conditional timing response on a nodal ray

For a weak thin local potential $\widehat{\psi}$, use the standard distance-matrix arrival-time function

$$\beta = D_{os}^{-1} X_s, \qquad T = (1+z_l) D_{ol}^T D_{ls}^{-1} D_{os},$$
$$ct(\theta) = \tfrac{1}{2}(\theta-\beta)^T T(\theta-\beta) - (1+z_l)\widehat{\psi}(D_{ol}\theta) + f(\beta). \tag{14}$$

This optical construction and its clock normalization are established distance-matrix lens theory [5, 6]. Put $D_{ab} = d_{ab}(I_2 + E_{ab})$ and $T_0 = (1+z_l) d_{ol} d_{os}/d_{ls}$. At first order $\delta T/T_0 = E_{ol} + E_{os} - E_{ls}$. For a stationary image, fixed measured redshifts, fixed transverse physical source X_s, and fixed local physical potential, the source variation is $\delta\beta = -E_{os}\beta$. Using the background lens equation and writing $v_i = \theta_i - \beta$,

$$\frac{c\delta t_i - \delta f}{T_0} = \tfrac{1}{2} v_i^T (E_{os} - E_{ol} - E_{ls}) v_i + v_i^T (E_{os} - E_{ol})\beta. \tag{15}$$

The expression includes the varying impact in the local potential. Image-position derivatives drop out by stationarity; the common f cancels in relative delays.

Choose $C_\star = \mathrm{diag}(1,-1,0)/\sqrt{2}$, the central ray $n = e_z$, and screen (e_x, e_y). Here Q and its full first gradient vanish on the ray, the transverse Hessian has zero trace, and $H_{C_\star} = \sqrt{2}\,\mathrm{diag}(1,-1)$. The first-order frequency, radial-event, velocity, local-scale and Jacobi trace responses therefore vanish. The first-order physical maps are fully specified by $E_{ab} = -\Gamma_{ab}$ for this preparation. These vanishing terms refer to the fiducial nodal ray. Actual images have small finite angular offsets; their leading tidal separation is included by the distance matrices. Higher finite-angle corrections are not bounded by this first-order paraxial calculation.

Take a local SIS, $\widehat{\psi}(x) = b|x|$, and a source $X_s = se$, $e = (\cos\varphi, \sin\varphi)$, with fixed $b, s > 0$. The background values are $\beta = s/d_{os}$, $\theta_E = b d_{ls}/d_{os}$, and $\theta_\pm = (\beta \pm \theta_E)e$. For $0 < \beta < \theta_E$ and fixed noncritical branches,

$$\Delta t_0 = t_- - t_+ = \frac{2(1+z_l) bs}{c} \frac{d_{ol}}{d_{os}}. \tag{16}$$

Since $v_\pm = \pm\theta_E e$, the quadratic term in Eq. (15) cancels between images:

$$\boxed{\frac{\delta\Delta t}{\Delta t_0} = e^T(\Gamma_{os} - \Gamma_{ol})e = \sqrt{2}[(g_{os} - g_{ol})\cdot y]\cos(2\varphi).} \tag{17}$$

The ls shear cancels for this fixed-physical-source SIS experiment. The result predicts the angular sign reversal and null directions for the declared preparation; it does not select the physical amplitude y. It is not a generic prediction for arbitrary galaxy profiles, fixed angular sources, finite-amplitude metric completions, or higher-order light propagation.

6 Executed conditional predictions

6.1 Six response rows

The constrained quadratic action is evolved from the six unit preparations in Eq. (3). A time-dependent linear map gives the two Newtonian metric potentials. The sum $\Phi + \Psi$ is then integrated separately for the ol, os and ls maps. This procedure preserves one common state through all segments while initializing each physical angle-to-separation map at its own vertex. Table 1 gives the resulting response coefficients at $(z_l, z_s) = (0.3, 1)$.

For example, the observer–source density-coordinate response is $g_{os,\delta_m} = -4.07274462 \times 10^{-4}$, while the observer–lens response is $-3.44476439 \times 10^{-5}$. Their difference produces $k_{\delta_m} = -5.27256743 \times 10^{-4}$. The nonzero result comes from the evolved environmental preparation and its integrated transverse Hessian; no measured delay is used to choose these coefficients.

Anchor	$g_{ol,j}$	$g_{os,j}$	$g_{ls,j}$	$k_j = \sqrt{2}(g_{os,j} - g_{ol,j})$
ζ	-7.589×10^{-6}	-5.650×10^{-5}	-2.236×10^{-5}	-6.918×10^{-5}
$\Pi_{2,\mathrm{ref}}$	6.054×10^{-5}	1.160×10^{-3}	6.427×10^{-4}	1.554×10^{-3}
δ_m^{ADM}	-3.445×10^{-5}	-4.073×10^{-4}	-2.042×10^{-4}	-5.273×10^{-4}
δ_r^{ADM}	-3.005×10^{-8}	-3.607×10^{-7}	-1.818×10^{-7}	-4.675×10^{-7}
v_m^{ADM}	3.009×10^{-4}	3.659×10^{-3}	1.853×10^{-3}	4.749×10^{-3}
v_r^{ADM}	3.612×10^{-7}	4.392×10^{-6}	2.224×10^{-6}	5.700×10^{-6}

Table 1: Executed unit-state coefficients at $(z_l, z_s) = (0.3, 1)$. The g rows determine fractional shear through Eq. (11); k_j determines the nodal SIS fractional delay at $\varphi = 0$. Initial coordinates and L^2 spatial normalization are fixed by Eqs. (3) and (5). These columns are response derivatives in that basis, not independently assigned physical perturbation sizes.

z_l	z_s	Δt_0 (days)	$\delta\Delta t(0°)$ (s)	$\delta\Delta t(45°)$ (s)	$\delta\Delta t(90°)$ (s)
0.3	1	7.753621	-0.353216	0	0.353216
0.3	0.6	9.268876	-0.123113	0	0.123113
0.5	1.5	11.583863	-1.163398	0	1.163398
0.8	2	17.264699	-2.871297	0	2.871297

Table 2: Conditional first-order delay predictions for the same normalized preparation, physical SIS strength and physical source displacement. The source azimuth is measured in the fixed nodal screen. The 45° values are exact first-order zeros; small nonzero values from a finite linear-map completion do not constitute second-order field predictions.

6.2 A finite illustrative preparation

Choose the nodal quadrupole $C_\star$ of Section 5 and

$$y_\star = (0, 0, 10^{-3}, 0, 0, 0)^T. \tag{18}$$

The number 10^{-3} is the coefficient of the normalized harmonic in the initial ADM matter-density contrast. It does not mean a spatially uniform 0.1% density perturbation. In particular, its initial spatial profile is $\delta_m^{\mathrm{ADM}}(\chi, n) = 10^{-3} Q_{C_\star}(\chi, n)$. Other components need not stay zero during the coupled evolution.

Set $\sigma_v = 200\,\mathrm{km\,s^{-1}}$, $b = 4\pi(\sigma_v/c)^2$, and $s = 0.8$ kpc. At $(z_l, z_s) = (0.3, 1)$ the background distances are

$$(d_{ol}, d_{os}, d_{ls}) = (945.162839, 1689.312240, 1076.994348)\ \mathrm{Mpc}. \tag{19}$$

The background source angle is 0.0976799 arcsec and the Einstein angle is 0.7354566 arcsec. The two unperturbed images on the chosen axis lie at +0.8331365 and −0.6377768 arcsec. Their baseline relative delay is 7.75362145 days. Equation (17) gives

$$\delta\Delta t = -0.35321609\cos(2\varphi)\ \mathrm{s} \quad [(z_l, z_s) = (0.3, 1),\ y = y_\star]. \tag{20}$$

The source rotation changes its orientation relative to the prepared quadrupole while keeping its physical displacement magnitude fixed. This is a family of controlled physical source configurations, not a rotation of an observed object with all other data held unchanged.

The larger-redshift examples give larger responses for this chosen state (Table 2); no universal monotonicity theorem is implied. Scaling $y_\star$ scales all first-order shifts together. Such a rescaling does not establish that the resulting amplitude is physically realized or measurable. At 2001 sampled radial points, exact angular maximization of the evolved seed gives maximum $2|\Phi| = 3.27 \times 10^{-6}$ through $z = 1$ and 2.04×10^{-5} through $z = 2$. The corresponding sampled first-order matter speeds are $4.89 \times 10^{-6}c$ and $1.72 \times 10^{-5}c$. These values support use of a weak perturbative illustration on the sampled domain. They are not certified continuous bounds or a calculation of nonlinear truncation errors.

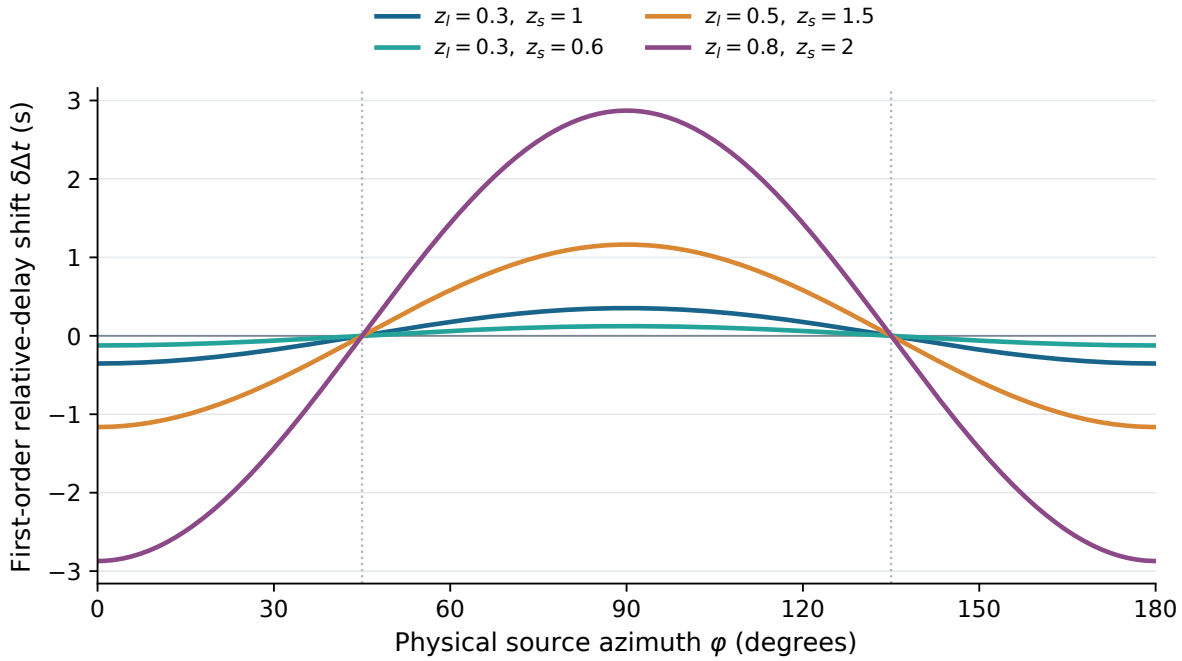


Figure 1: The calculated angular delay law for the four configurations in Table 2. Every curve uses the single specified preparation in Eq. (18). The phase, null directions and sign reversal follow from the common quadrupolar shape; the amplitudes follow from the six-state propagation and background distances. The curves are theoretical forward responses.

6.3 Shared-state relations

Equation (13) imposes collinearity of the smooth shear components whenever all six coordinates share one spatial tensor C. An explicit calculation at a generic line of sight and generic normalized C produces pairwise two-component shear determinants below 1.01×10^{-28} over the twelve reported propagation segments. This is a numerical check of the algebraic common-profile identity. It is not an additional independent constraint on arbitrary image ellipticities or a statement about noise covariance.

The general preparation contains six independent coefficient sets in each of five quadrupole directions. For that larger family the useful restriction is the common temporal response design $S_{rA} = G_{rj} y_{jA}$. Angular information alone does not select the six temporal amplitudes. Any proposed inference must also evaluate the redshift design, its conditioning and the allowed physical nuisance parameters.

7 Effective gravitational representation and source-redshift dependence

The computed optical response admits a conventional effective-potential representation. This correspondence is constructive: the geometric and effective gravitational descriptions encode the same retained light transfer. The existence of that representation does not identify the physical preparation from lensing data. Distance-matrix lensing and its associated degeneracies provide the appropriate comparison [5–7].

7.1 Exact correspondence for one source plane

Use physical angle-to-separation maps in compatible parallel screen frames, constant across the paraxial beam. Let T be the symmetric time-delay matrix, $T_0 > 0$ a scalar reference length, and $M = T/T_0$. The local physical lens potential $\hat{\psi}$ and the angular source coordinate $\boldsymbol{\beta} = D_{os}^{-1}\boldsymbol{X}_s$ give

$$ct(\boldsymbol{\theta}) = \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\beta})^T T(\boldsymbol{\theta} - \boldsymbol{\beta}) - (1 + z_l)\hat{\psi}(D_{ol}\boldsymbol{\theta}) + f(\boldsymbol{\beta}). \tag{21}$$

The arbitrary source-only function f cancels between images of the same source. Define

$$\boldsymbol{\beta}_{\text{eff}} = M\boldsymbol{\beta}, \tag{22}$$

$$\psi_{\text{eff}}(\boldsymbol{\theta}) = \frac{1 + z_l}{T_0}\hat{\psi}(D_{ol}\boldsymbol{\theta}) + \frac{1}{2}\boldsymbol{\theta}^T(I - M)\boldsymbol{\theta}. \tag{23}$$

Expansion of the quadratic terms gives the exact identity

$$\begin{aligned} \frac{ct(\boldsymbol{\theta})}{T_0} &= \frac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}_{\text{eff}}|^2 - \psi_{\text{eff}}(\boldsymbol{\theta}) + F_s, \\ F_s &= \frac{1}{2}\boldsymbol{\beta}^T(M - M^2)\boldsymbol{\beta} + \frac{f(\boldsymbol{\beta})}{T_0}. \end{aligned} \tag{24}$$

Since F_s is independent of image position, the stationary images and relative arrival times are identical in the two representations. The effective deflection is

$$\boldsymbol{\nabla}\psi_{\text{eff}} = \frac{1 + z_l}{T_0}D_{ol}^T\hat{\boldsymbol{\alpha}}(D_{ol}\boldsymbol{\theta}) + (I - M)\boldsymbol{\theta}. \tag{25}$$

This is an application of the established arbitrary-background Fermat formalism [5]; the state-dependent matrices evaluated here supply its specified cosmological realization.

For the physical SIS potential $\hat{\psi}(\boldsymbol{x}) = b|\boldsymbol{x}|$, the first term in Eq. (23) is an elliptical potential when D_{ol} is anisotropic. The quadratic term is a uniform-convergence and shear contribution. In the nodal quadrupole its first-order correction is tracefree. Agreement with this effective model therefore follows from the correspondence and is not separate evidence for the underlying state. Neither a globally admissible mass distribution nor agreement with independent dynamical constraints follows from the algebra alone. Source-position and mass-sheet freedoms must be included when assessing what an observation can determine [7].

7.2 A restricted comparison across two source redshifts

Consider sources s and t behind the same lens, using the same D_{ol}, physical lens potential, screen frame and small angular patch. For each source, define background quantities

$$\begin{aligned} p_s &= \frac{d_{ls}}{d_{os}}, & T_{0,s} &= (1 + z_l)\frac{d_{ol}}{p_s}, \\ a_s &= \frac{1 + z_l}{T_{0,s}} = \frac{p_s}{d_{ol}}, & B_s &= I - \frac{T_s}{T_{0,s}}. \end{aligned} \tag{26}$$

The equivalent potential has the form $\psi_{\text{eff},s} = a_s\hat{\psi}(D_{ol}\boldsymbol{\theta}) + \boldsymbol{\theta}^T B_s\boldsymbol{\theta}/2$. An additional physical quadratic deflector at the main lens, $\delta\hat{\psi}(\boldsymbol{x}) = \boldsymbol{x}^T A_{\text{lens}}\boldsymbol{x}/2$, can absorb all these quadratic terms with one fixed symmetric matrix A_{lens} precisely when

$$\frac{B_s}{a_s} = D_{ol}^T A_{\text{lens}} D_{ol} \quad \text{is independent of } s. \tag{27}$$

For common nonsingular D_{ol} this is equivalent to source independence of B_s/p_s. To first order about $D_{ol} = d_{ol}I$, it is the familiar physical-to-reduced scaling $B_s = p_s d_{ol} A_{\text{lens}}$.

The tracefree contrast appropriate to the present calculation is

$$R_{st} = \frac{B_s^{\mathrm{TF}}}{p_s} - \frac{B_t^{\mathrm{TF}}}{p_t}. \tag{28}$$

With $E_{ab} = -\Gamma_{ab}$ and the common quadrupolar state y,

$$\begin{aligned} B_s^{\mathrm{TF}} &= [(g_{ol} - g_{ls} + g_{os}) \cdot y] H_C + O(|y|^2), \\ R_{st} &= r_{st} H_C + O(|y|^2), \\ r_{st} &= \left[\frac{g_{ol} - g_{ls} + g_{os}}{p_s} - \frac{g_{ol} - g_{lt} + g_{ot}}{p_t} \right] \cdot y. \end{aligned} \tag{29}$$

A thin tidal forcing exactly at the main-lens position provides a useful check: $g_{ol} = g_{ls} = 0$ and $g_{os} \propto S_K(\chi_l) S_K(\chi_s - \chi_l)/S_K(\chi_s) = S_K(\chi_l) p_s$. Its contrast vanishes, whereas a distributed response need not obey that scaling.

For the specified nodal tensor $C_\star = \mathrm{diag}(1, -1, 0)/\sqrt{2}$, the illustrative state with $y_{\delta_m} = 10^{-3}$ and all other initial amplitudes zero gives, at $z_l = 0.3$, $z_s = 1$ and $z_t = 0.6$,

$$R_{st} = -9.2059866 \times 10^{-8} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{at first order.} \tag{30}$$

This dimensionless response cannot be absorbed into the one fixed physical main-plane quadratic term specified above. It remains compatible with a distributed conventional gravitational representation of the same light transfer. The coefficients B_s are model quantities whose inference requires source, lens and measurement modeling; Eq. (30) is not a detected astronomical residual.

In particular, the contrast does not establish general degeneracy breaking. Nonquadratic lens freedom, source-position transformations, additional lens planes and uncertain distance factors require their own joint treatment. Generalized mass-sheet transformations can scale delays for multiple source planes together [6]. The result here is narrower: one shared preparation predicts a definite redshift dependence of its effective quadratic contribution, and that dependence is constrained when the physical deflector is required to be shared.

8 Verification and reproducibility

The package separates inherited inputs, new analytic identities, executed response arrays, lens applications and independent checks. The inherited source files are exact blobs from a named commit; their Git identities, SHA-256 hashes and byte lengths are recorded. The historical manifest is retained for provenance, but only its frozen background fields enter this calculation. Historical observational entries in that file are inert.

The main temporal integration uses 801 interpolation points, with a 1601-point refinement and tighter integration tolerances. Each Green-function shear integral is compared with a separately integrated forced Jacobi equation. A further check integrates the original angular Euler momentum equation on the nodal geometry, rather than reusing the Green kernel. The angle and momentum variables satisfy

$$\theta' = \frac{L}{S_K^2(\chi)}, \qquad L' = -w(\eta_o - \chi)\mathcal{N} \sin^2(\chi/R)\, h\theta, \qquad h = \pm\sqrt{2}, \tag{31}$$

with the appropriate independently initialized vertex conditions. Here $w = \sum_j (\phi_j + \psi_j) y_j$ is the temporal Weyl coefficient for the prepared state, and primes denote $d/d\chi$. Complex-step and real finite-difference derivatives agree with the kernel result. The apparent coordinate singularity at the observer is handled through the regular background initialization described in the executable audit.

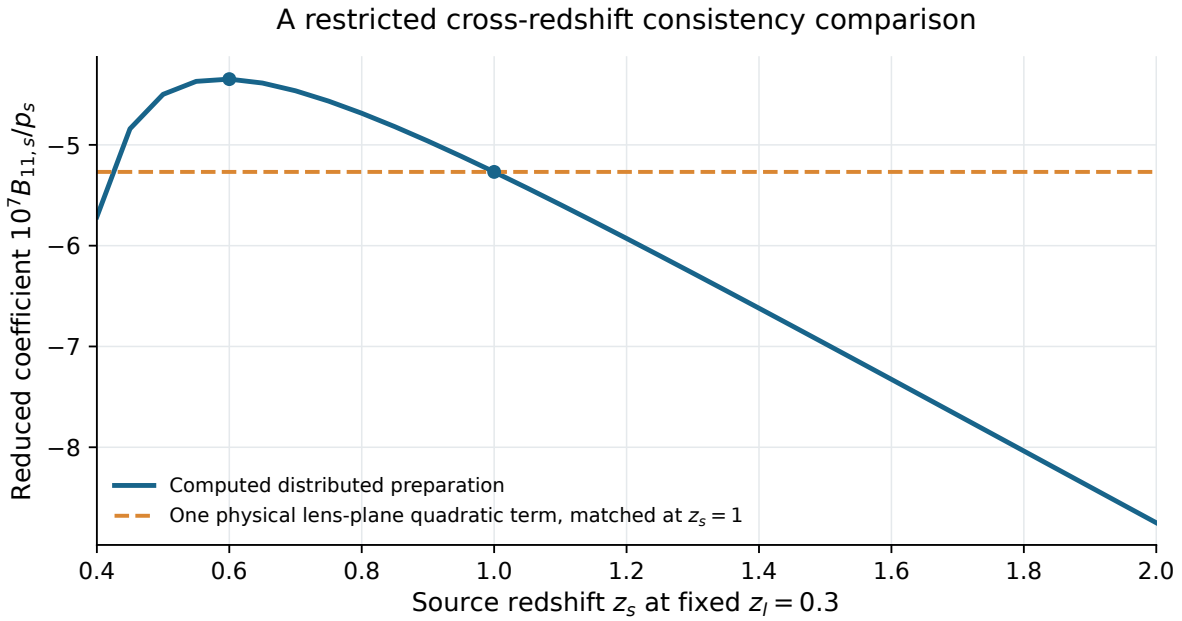


Figure 2: The calculated first-order nodal coefficient $B_{11,s}/p_s$ for the same normalized density preparation and $z_l = 0.3$, sampled at 33 source redshifts. The dashed comparison is a single fixed physical quadratic potential at the main lens, matched at $z_s = 1$. The contrast tests that restricted absorption criterion. It is not a direct observable or an exclusion of distributed gravitational structure, other lens profiles or generalized mass-sheet freedom.

Check	Maximum discrepancy
S^3 harmonic Gram matrix versus I_9 (volume quadrature)	3.49×10^{-14}
Six physical anchors reconstructed from the canonical initial state	2.22×10^{-16}
Original action constraints, scaled by $\max(1, \|\boldsymbol{w}\|_\infty)$	1.46×10^{-11}
Green kernel versus forced Jacobi equation, absolute coefficient	1.45×10^{-14}
Refined temporal integration, absolute kernel coefficient	1.47×10^{-13}
Original angular-ray derivative versus kernel, complex-step	2.08×10^{-15}
Original angular-ray derivative, real half-step comparison	5.70×10^{-13}
Full image-solve fractional delay derivative versus analytic law	7.12×10^{-12}
SI baseline reconstruction, relative delay discrepancy	6.24×10^{-16}
Effective-potential Fermat/gradient identity, scaled	2.01×10^{-16}

Table 3: Executed checks of the declared equations and their numerical application. The rows use the indicated normalizations; their values are not interchangeable observational uncertainties or bounds on omitted physical terms.

The strong-lensing calculation solves the two noncaustic image branches and evaluates the full matrix Fermat function. Four-point derivatives in independent map perturbations reproduce the stationary-image formula. A separate check reconstructs the SIS baseline directly in SI units and retains the time-matrix, source-angle and physical-impact variations before simplification. The effective-potential identity is checked both in its value, up to a source-only constant, and in its angular gradient.

The finite-map experiment sets $D_{ab} = d_{ab}(I - \Gamma_{ab})$ at a small finite input and recomputes the images. It checks the derivative of that map construction. Products generated by this completion are not a second-order solution of Eq. (1). In particular, its nonzero residual at the 45° first-order null is not promoted to a new physical prediction. Numerical precision is kept distinct from the approximation order and physical preparation.

9 Scientific interpretation and limits

The forward calculation establishes a concrete connection between the state and two kinds of light response. The spatial harmonic determines the screen tensor, while the same six-state temporal evolution fixes the segment-dependent amplitudes. The SIS experiment then reduces a full stationary arrival-time variation to a difference of two such amplitudes. The angular orientation of the quadrupole, its timing amplitude and its shear pattern cannot be assigned independently within this declared preparation.

The special nodal geometry makes the result particularly transparent. The fiducial light ray has no first-order scalar frequency or radial shift, yet the neighboring rays feel a nonzero transverse tidal field. This permits a complete first-order relative-delay calculation within the specified paraxial experiment. A generic sightline would require the trace, endpoint and frequency responses as well. The tracefree kernel in Eq. (12) remains useful there, but it cannot stand in for the missing scalar distance calculation.

There are several distinct levels of physical specification. The effective action and homogeneous background are fixed inputs. The initial vector and spatial profile are explicit conditional choices. The SIS is a prescribed local physical encounter. No observed amplitude, unique microscopic envelope state, stable nonlinear completion or population-level distribution of quadrupole orientations is inferred. Each of these would add physical content beyond the calculations reported here.

The representation result explains why a successful forward calculation need not appear as an extra contribution when expressed in familiar lens variables. Its effective potential already contains the propagated response. Adding an independent copy of the same surface contribution would double count it. The restricted redshift contrast gives a way to state one common-physical-lens condition precisely, while leaving the broader class of distributed gravitational representations available.

Observational identification is a separate problem. Locally, let

$$\delta\mathcal{O} = \mathsf{B}\,\delta y + \mathsf{N}\,\delta\eta + \epsilon, \qquad \mathsf{B}_\perp = (I - P_{W\mathsf{N}})W\mathsf{B}, \qquad W^T W = \mathsf{C}^{-1}. \tag{32}$$

Here C is the observational covariance, η denotes the allowed lens, source and environmental nuisance parameters, and $P_{W\mathsf{N}}$ projects onto their whitened response columns. The rank and conditioning of $\mathsf{B}_\perp$ determine which local state directions survive those freedoms. No such data covariance or fully specified observational nuisance design is supplied by the present forward example, so it supplies no detection significance or achievable parameter error. Generalized mass-sheet transformations can rescale relative delays together; multiple source redshifts or delay ratios alone do not guarantee identification [6].

A useful next observational design would specify one common preparation for several source redshifts, a physically constrained main lens and line-of-sight environment, and independent information on the source or lens dynamics. The response rows calculated here would then be inputs to that design. Giving every image or source its own unconstrained initial state would remove the shared-state relation that motivates the calculation.

10 Conclusions

The full scalar $n = 2$ family of the declared closed BHSM/R1 realization supports an explicit angular optical calculation. Its nine spatial modes decompose into $1 + 3 + 5$ sky sectors. The four monopole/dipole directions are absent from first-order smooth shear, while the five quadrupoles have a simple normalized transverse Hessian. Combining that Hessian with the six physical temporal responses gives the tracefree parts of the three propagation maps. On the specified nodal ray these also give the full first-order maps for the retained lens experiment.

For one common nodal quadrupole and a fixed physical SIS/source preparation, the delay obeys $\delta\Delta t/\Delta t_0 = \sqrt{2}[(g_{os} - g_{ol}) \cdot y]\cos(2\varphi)$. The computed example gives a 0.353216 s amplitude at $(z_l, z_s) = (0.3, 1)$, with a specified sign, 90° sign reversal and 45° first-order null. Independent propagation, image solving and dimensional reconstruction check the result.

An exact effective-potential correspondence preserves image equations and relative arrival times for one source plane. A calculated two-redshift contrast tests absorption into one fixed physical quadratic term at the main lens, without claiming general degeneracy breaking. The completed contribution is a reproducible family of conditional BHSM light-transfer and timing predictions, with the state selection and observational inference left as explicitly separate physical tasks.

A Frozen effective action and six-state temporal provider

This appendix specifies the conditional R1 realization used for the temporal response coefficients. It retains the coupled scalar, dust, and radiation system of the environmental and transport source calculations [1, 2]. It does not infer the microscopic BHSM state or its absolute normalization. The vendored source files and their hashes identify the precise implementation; only the background entries of the inherited parameter manifest are used.

A.1 Action, units, and background

The metric signature is $(-,+,+,+)$ and dots denote cosmic-time derivatives. In the code units $H_0 = M_{\rm Pl} = c = 1$, the effective action, up to its constant dimensional prefactor, is

$$S = \int d^4x\sqrt{-g}\left[\frac{\mathcal{R}}{2} + X - V(T) - \frac{2X}{\lambda}\Box T\right] + S_m + S_r, \qquad X = -\frac{1}{2}\nabla_\mu T\nabla^\mu T, \quad V(T) = V_0 - q_T T. \tag{33}$$

Thus the cubic term is $+(\nabla T)^2\Box T/\lambda$, with no additional factor of 1/2. In the convention with dimensional cubic scale Λ, $G_3 = 2X/\Lambda^3$ and $\lambda = \Lambda^3/(M_{\rm Pl}H_0^2)$ after rescaling $T_{\rm phys} = M_{\rm Pl}T$ and time by H_0^{-1}. The fluids are minimally coupled, isentropic, irrotational perfect fluids with $w_m = 0$ and $w_r = 1/3$; the dust sound speed is exactly zero.

The frozen parameters are

$$\begin{gathered}(\Omega_{m0}, \Omega_{r0}, \Omega_{k0}, \lambda, q_T) = (0.31, 9\times10^{-5}, -0.018, 1, 3),\\ V_0 = 2.3986073449708067.\end{gathered} \tag{34}$$

Here $K = -\Omega_{k0} = 0.018$ is the spatial curvature in H_0^2/c^2 units; V_0 is in $M_{\rm Pl}^2H_0^2$ units. With $v = \dot{T}$, the background implementation uses

$$\begin{aligned} J &= v + 6Hv^2/\lambda, & \dot{J} + 3HJ &= q_T,\\ \rho_T &= \tfrac{1}{2}v^2 + V + 6Hv^3/\lambda, & 3(H^2 + K/a^2) &= \rho_m + \rho_r + \rho_T, \end{aligned} \tag{35}$$

where $\rho_m = 3\Omega_{m0}a^{-3}$ and $\rho_r = 3\Omega_{r0}a^{-4}$. At $z = 100$, the inherited background prescription sets $T = 0$ and $J = 2q_T/(9H)$, selecting the positive-v solution jointly with the Friedmann constraint. The stored V_0 comes from the original $H(a = 1) = 1$ normalization; this calculation neither repeats that shooting nor retunes it. Replaying the frozen input gives $H(a = 1) = 0.9999999999811999$, a normalization residual of -1.88×10^{-11}. The background is integrated in $N = \ln a$ with DOP853, relative tolerance 2×10^{-13}, absolute tolerance 10^{-15}, and maximum step 0.005.

A.2 Finite harmonic action and reduction

For one real harmonic, $\int Q^2 dV_\gamma = 1$ and $\nabla_\gamma^2 Q = -8KQ$. The scalar-field uniform-T ADM convention uses lapse perturbation α, covariant shift $N_i = \nabla_i(\chi_{\rm sh}Q)$, and spatial curvature

perturbation ζ. All variables below are harmonic coefficients. Define

$$\begin{gathered}\boldsymbol{x}=(\zeta,D_m,D_r)^T,\qquad D_I=\frac{\delta\rho_I}{\rho_I+p_I}+3\zeta,\qquad \boldsymbol{u}=(\alpha,\chi_{\rm sh},v_m,v_r)^T,\\ \kappa=K/a^2,\qquad b_2=8\kappa,\qquad s_2=5\kappa.\end{gathered}\tag{36}$$

The symbols v_m, v_r here denote ADM fluid velocity potentials, not the background field velocity v. Put $h_I=\rho_I+p_I$ and

$$\Theta=H-v^3/\lambda,\qquad \Sigma=\tfrac{1}{2}v^2+12Hv^3/\lambda-3H^2.\tag{37}$$

After algebraically eliminating the spatial fluid-current auxiliaries and performing the declared integration by parts, the implemented polynomial is

$$\begin{aligned}\frac{L_2}{a^3}=&-3\dot\zeta^2+s_2\zeta^2+\Sigma\alpha^2+2b_2\Theta\alpha\chi_{\rm sh}-2b_2\dot\zeta\chi_{\rm sh}\\&+6\Theta\alpha\dot\zeta+2s_2\alpha\zeta-b_2\kappa\chi_{\rm sh}^2+\sum_{I=m,r}h_I\ell_I,\\ \ell_I=&\,v_I\dot D_I-\frac{b_2}{2}v_I^2+b_2\chi_{\rm sh}v_I-\frac{w_I}{2}(D_I-3\zeta)^2-\alpha(D_I-3\zeta).\end{aligned}\tag{38}$$

This equation specifies every finite Hessian entry used by the provider. Write it as

$$L_2/a^3=\tfrac{1}{2}\boldsymbol{u}^TA\boldsymbol{u}+\boldsymbol{u}^T(B\dot{\boldsymbol{x}}+D\boldsymbol{x})+\tfrac{1}{2}\dot{\boldsymbol{x}}^TM_0\dot{\boldsymbol{x}}+\tfrac{1}{2}\boldsymbol{x}^TE\boldsymbol{x}.\tag{39}$$

The auxiliary equations and their Schur reduction give

$$\begin{aligned}&\boldsymbol{u}=-A^{-1}(B\dot{\boldsymbol{x}}+D\boldsymbol{x}),\\ &M=M_0-B^TA^{-1}B,\qquad L=-B^TA^{-1}D,\qquad V_2=E-D^TA^{-1}D.\end{aligned}\tag{40}$$

The reduced canonical momentum is $\boldsymbol{p}=a^3(M\dot{\boldsymbol{x}}+L\boldsymbol{x})$. Consequently the six-dimensional state $\boldsymbol{w}=(\boldsymbol{x},\boldsymbol{p})^T$ evolves by

$$\dot{\boldsymbol{x}}=M^{-1}(a^{-3}\boldsymbol{p}-L\boldsymbol{x}),\qquad \dot{\boldsymbol{p}}=a^3(L^T\dot{\boldsymbol{x}}+V_2\boldsymbol{x}).\tag{41}$$

Dividing these equations by H defines $d\boldsymbol{w}/dN=F(N)\boldsymbol{w}$ and the full fundamental matrix $U'=FU$, $U(N_i)=I_6$. No projection or restart onto an isolated two-state sector is performed. The default propagator tolerances are $(2\times10^{-10},2\times10^{-12})$, with maximum N step 0.01; the refinement uses $(2\times10^{-12},2\times10^{-14})$ and step 0.005.

A.3 Physical anchor and optical potential rows

At $z_i=2.1$, $N_i=-\ln 3.1$, the six specified coordinates are

$$\begin{aligned}&\boldsymbol{y}=(\zeta,\Pi_{2,\rm ref},\delta_m^{\rm ADM},\delta_r^{\rm ADM},v_m^{\rm ADM},v_r^{\rm ADM})_i^T,\\ &\Pi_{2,\rm ref}=2a_i^3G_{S,2}^{\rm ref}\dot\zeta_i,\qquad G_{S,2}^{\rm ref}=\frac{5(r+3)}{r+8},\qquad r=\frac{\Sigma}{\Theta^2}.\end{aligned}\tag{42}$$

The reference factor is the inherited gravity–scalar normalization; $\Pi_{2,\rm ref}$ is not identified with the coupled canonical momentum p_ζ. The current action rows implement $\delta_m^{\rm ADM}=D_m-3\zeta$, $\delta_r^{\rm ADM}=\frac{4}{3}(D_r-3\zeta)$, and the reconstructed velocity potentials. If $\mathcal{B}_i\boldsymbol{w}_i=\boldsymbol{y}$ denotes these six rows, the initial canonical state is obtained by solving this full system, $\boldsymbol{w}_i=\mathcal{B}_i^{-1}\boldsymbol{y}$. All six coordinates are retained independently for every spatial harmonic.

The Newtonian metric convention is $ds^2=-(1+2\Phi)dt^2+a^2(1-2\Psi)\gamma_{ij}dx^idx^j$. For one harmonic, $\Phi=\phi Q$ and $\Psi=\psi Q$, with reconstructed rows

$$\begin{aligned}&\phi=\alpha+\dot\chi_{\rm sh},\qquad \psi=-\zeta-H\chi_{\rm sh},\\ &\delta_I^{\rm N}=(1+w_I)(D_I-3\zeta-3H\chi_{\rm sh}),\qquad v_I^{\rm N}=v_I^{\rm ADM}-\chi_{\rm sh}.\end{aligned}\tag{43}$$

The derivative of χ_{sh} includes both the background dependence of its reconstruction row and the propagated state. The code evaluates the former by a complex-step directional derivative. Multiplication of these metric rows by $U(N, N_i)\mathcal{B}_i^{-1}$ gives the six columns $\phi_j(N), \psi_j(N)$ used by the optical integrals. Cubic interpolation uses 801 temporal samples, repeated with 1601 samples in the refinement. The reported reconstruction, action-constraint, and independent optical checks establish numerical consistency of this declared finite system. They do not establish a general stability theorem, select its initial state, or supply the missing microscopic BHSM preparation.

B Reproduction and source provenance

The accompanying package contains the source, executed JSON receipts, temporal arrays, exact-identity checks, figures and this manuscript. The root `README.md` gives a single replay command and the individual stages. Python 3.12.14, NumPy 2.3.5, SciPy 1.17.0 and Matplotlib 3.10.8 were used. Reproduction reads no remote data and requires no telescope archive access.

The inherited action/background/anchor provider is frozen at commit `06cf7c1` of the *Manuscript-Generation* repository. The corresponding environmental derivation is referenced at commit `00614fa0` of the *Berger-Hopf-Standard-Model* repository. The new angular kernels, nodal timing application, shared-state contrast and manuscript are packaged separately. SHA-256 manifests allow the inherited inputs and new release files to be distinguished.

Author and disclosure statements for review

Author contribution. Norman P. Carberry: conceptual direction and authorship. The author is responsible for review and approval of the submitted scientific claims.

Computational assistance. AI-assisted tools were used to develop code, draft mathematical exposition and perform separate implementation reviews. Analytic identities, independent numerical formulations and executable receipts are supplied so that claims can be checked without relying on those tools' judgments.

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Data availability. All numerical results in this draft are synthetic forward calculations from the declared inputs. Their source and receipts are included in the accompanying reproducibility package. No human-subject or observational target data were analyzed.

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